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A New Index of Occupational Exposure

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ABSTRACT¹

This paper introduces the AI Generated Index of Occupational Exposure (GENOE), a novel measure quantifying the potential impact of artificial intelligence on occupations and their associated tasks. Our methodology employs synthetic AI surveys, leveraging large language models to conduct expert-like assessments. This approach allows for a more holistic evaluation of job replacement likelihood, reducing assumptions about the mechanisms through which AI innovations could replace job tasks and skills. Our findings reveal that the average occupational exposure to AI is 0.28 within one year, increasing to 0.38 and 0.44 over the five- and 10-year horizons, respectively. We also show evidence that the index not only considers task automation, but also contextual factors such as social and ethical considerations and regulatory constraints that may affect the likelihood of replacement. After calibrating the index with labor market microdata from the United States and Mexico, we show that approximately 43 million and 16 million jobs, respectively, are highly exposed to AI within a one-year time horizon. The GENOE index provides valuable insights for policymakers, employers, and workers, offering a data-driven foundation for strategic workforce planning and adaptation in the face of rapid technological change.

JEL classifications: C53, C81, J23, J24, O33

Keywords: Artificial intelligence (AI), Labor markets, Job displacement, Automation, Occupational exposure, Tasks, Synthetic AI surveys, AI Generated Index (GENOE)

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